



afeef ahmed <afeefahmed.shaik@gmail.com>

Third International Conference on computational Intelligence and Computing Applications : Submission (82) has been created.

1 message

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Tue, Apr 21, 2026 at 3:31 PM

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The following submission has been created.

Track Name: ICCICA2027

Paper ID: 82

Paper Title: Multi-Criteria Refugee-to-City Allocation via Mixed-Integer Linear Programming: A Scalable Decision-Support Framework

Abstract:

Refugee resettlement poses a difficult resource allocation challenge involving the consideration of various socio-economic and humanitarian issues in light of hard resource limitations. In this paper, we develop a flexible and extensible optimization-driven support tool based on Mixed Integer Linear Programming to optimize the allocation of refugees to their host cities based on an eight-parameter compatibility framework including language skills, educational background, cost preferences, employment opportunities, diaspora support, policy considerations, trauma sensitivity, and other parameters. Refugees are first partitioned into homogeneous groups to facilitate optimization scalability, and the process is implemented using the CBC optimizer via Google OR-Tools under four policy scenarios, namely Balanced, Capacity Stress, Jobs Priority, and Integration Priority. Our experimental evaluation on a synthetic data set comprising 50,000 refugees and 50 cities indicates that our solution achieves up to 14.2% performance improvement compared to a greedy algorithm baseline in terms of the objective score, better coverage of refugees, and a considerable reduction in the number of unassigned refugees. Finally, we integrate our approach into a RESTful API interface for policymakers' use.

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Authors:- afeefahmed.shaik@gmail.com (Primary)

Secondary Subject Areas: Not Entered

Submission Files:

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Submission Questions Response: Not Entered

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